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Aeginetia indica L. : An interesting root parasite of Dalma Wildlife Sanctuary, Jharkhand

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Abstract : *Aeginetia indica* L. is a root holoparasitic herb of family Orobanchaceae. This paper dealswith the distribution, ecology and conservation status of the *A indica* L. It is a threatened species of Dalma Wildlife Sanctuary. This species is under threat because of habitat loss and climate change.

Keywords : *Aeginetia indica* L., root holoparasite, conservation status, threatened species

INTRODUCTION

Aeginetia indica L.[1] is a small genus of family Orobanchaceae (broomraph family). It is a leaflessannual herb with beautiful and attractive purple flowers. It is a root holoparasite which uses grasses as hosts[2]. In India, family Orobanchaceae is represented by following genera: *Aeginetia, Boschniakia, Campbellia, Cistanche, Orobanche, Xylanche*[3] from which *Aeginetia* and *Orobanche* are reported in Jharkhand[4]. Haines, H.H. (1910, 1992, and 1925) in his books "A Forest flora of Chotanagpur including Gangpur and the Santal Parganahs" [5] and"The Botany of Bihar and Orissa"[6] published during the early years of 20th century are the pioneer floristic accounts that include the parasitic angiosperms of the family Orobanchaceae in Jharkhand.Someobservations of the root parasitic genus *Aeginetia indica* Linn.had done by Chavan et. al[7]. (1961).Phoenix and Press (2005)[8] have linked the physiological traits to impacts on community structure and function and the role of this roothemiparasitic Orobanchaceae.They also reviewed awide range of possible effects from climatechange on parasitic plants and on thecommunities in which they occur.Sharma and Uniyal (2009)[9] have reported *Aeginetia indica*L. as a new record to the flora of Himachal Pradesh on the host *Dendrocalamus strictus*.

The secondary metabolites present in *Aeginetiaindica* L. is aeginetic acid, aeginetolide and apocarotenoid polyenes [10]. The whole plant is used as folk medicine in infusions as a febrifuge to treat chronic liver diseases, cough and arthritis [11]. Ithas traditionally been used as a tonic and an anti-inflammatory medicinal plant. It has been used for the treatment of many diseases such as hepatitis and liver diseases. *Aeginetia indica Roxb*. has used in traditional medicine for the treatment diabetes and dermal swelling[12]. Anti-hepatotoxicity principles of *Aeginetia indica* is used to treat various ailments of liver[13]. Protein isolated from *Aeginetia indica* L. has antitumor effect which is used to treat tumors [14] [15]. *Aeginetia indicaused* as novel alternative therapeutic agents in the treatment of human renal cancer[16].

MATERIALAND METHODS

During the botanical explorations in Pindrabera forest area of Dalma Wildlife Sanctuary in 2012 some specimens of A. indica L. were collected at the steep valley of the core forest area at 1500- 2000 feet on the forest floor of Sal forest. Herbarium was prepared and preserved.

OBSERVATION

Specimens were observed and identified using different district floras of Botanical Survey of India [6] [17] whichare described below.

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***Aeginetia indica* L., Sp. Pl. 632. 1753; Hook. f., Fl. Brit. India 4: 320. 1884; Haines, Bot. Bihar Orissa 642. 1922 (Repr. ed., 2: 673. 1961).**

Description: Rootparasitic herb. Rhizome short, simple, thick, and irregularly cylindrical. **Roots:** few, wiry, thick. **Scape:**solitary, 15-35 cm long. Glabrous, decurved. **Calyx:** spathaceous, pinkish, 1-5cm long, acute. **Corolla:** lobes suborbicular- reniform, purple,margines fimbricate. **Androecium:** Stamens 4, didynamous; filaments glabrous; anther cells divergent, one sterile and one fertile, anther of lower stamens with a thick, gibbous, obtuse, spur behind. **Gynoecium:** Ovary of two carpels, syncarpous, unilocular; placentation parietal; style glabrous, persistent; stigma large, peltate. **Fruit:** Capsules 2- valved, fleshy, and irregularly dehiscing.

Distribution: India, China, Sri Lanka, Japan and Philippines

Flowering & Fruiting: July-November.

Ecology: Under thick shades of Sal forests as a root parasitic herb.



Fig 1: Habit of *Aeginetia indica* L.

RESULT & DISCUSSION

The family Orobanchaceae has only one representative member of root holoparasitic group reported by the author. This family is represented by the genus Aeginetia and species *Aeginetia indica*L.

Aeginetiais a small genus comprising three species in which only *Aeginetiaindica* L. is reported in Dalma. The calyx was fairly dominant, but the outer part of the corolla emerges through a slit on one side of the calyx as in *Aeginetia indica*L. There are some variations in the flower colours, and a form with a beautifully striped calyx is present. *Aeginetiaindica* L. also has a sterile trigger spur from the anther. One capsule can produce up to thousands seeds. The holoparasites are generally more host specific.*Aeginetiaindica* L. mostly attacks monocotyledons.

During the present investigations, this parasitic plant species was located growing in the community of Sal and its associates comprising *Anogeissus latifolia, Buchnania lanzen, Croton roxburghii, Diospyros melanoxylon, Haldinia cordifolia, Lagerstroemia parviflora, Lannea coromandelica, Madhuca longifolia, Oroxylum indicum, Pterocarpus marsupium, Scleichera oleosa, Semecarpus anacardium, Syzygium cumini, Sterculia urens, Terminalia alata, Terminalia bellirica, Terminalia chebula.* The species was found in Pindrabera, Manjhla bandh in the slopes of the Sal forest. There is no information about its abundance and distribution in Dalma Wildlife Sanctuary, where it is locally restricted only at higher altitudes.

The accessibility of these plants at higher altitude may assist in species protection but there are visible traces of influence of wild elephants, forest fires and forest clearing for agriculture and climate change due to anthropogenic activities. Therefore, this small population requires immediate conservation consideration by the appropriate national and local authorities because it is locally threatened.

CONCLUSION

A. indica L. is a threatened root parasitic angiosperm of Dalma Wildlife Sanctuary which has a lot of ethnomedicinal importance. The whole plant is used as folk medicine in infusions as a febrifuge to treat chronic liver diseases, cough and arthritis.*A. indica* L. has traditionally been used as a tonic and an anti-inflammatory medicinal plant.It has been used in traditional medicine for the treatment of several diseases including diabetes and dermal swelling.

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